

A New Impedance Measurement System for PZT-Based Structural Health Monitoring

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Abstract—Recent research has shown that electromechanical (E/M) impedance-based techniques are an efficient strategy for damage sensing in the context of structural health monitoring (SHM). The basic principle is to excite the lead zirconate titanate (PZT) sensors attached to the structure and to measure the corresponding electrical impedance. A variation in the electrical impedance indicates a variation in the mechanical impedance and, consequently, structural damage. To measure the electrical impedance, most universities and research centers use expensive, bulky, heavy, and high-powered impedance analyzers, which limit new developments and practical applications. To overcome these issues, this paper presents an efficient and inexpensive methodology for electrical impedance measurement. To verify the efficiency and accuracy of the new methodology, practical tests were carried out, and the results were compared with those obtained using a conventional impedance analyzer.

Index Terms—Impedance analyzer, smart structures, structural health monitoring (SHM), virtual instrumentation.

I. INTRODUCTION

THE FIELD OF structural health monitoring (SHM) has received attention from not only the academic community but also the private industry. The use of SHM to detect damages in concrete structures, such as bridges and large buildings, or in mechanical structures is something recent and promising. The objective of any application is to offer a high degree of security and to significantly reduce the cost of maintenance [1]–[3]. There are different methodologies to develop an SHM system [4]–[6], but special attention is given to those based on electromechanical (E/M) impedance obtained from piezoelectric transducers, such as lead zirconate titanate (PZT) ceramics, bonded in the structure to be monitored. The main characteristic of these transducers is the piezoelectric effect. From this property, there is a direct relationship between electrical impedance and mechanical impedance. Consequently, a structural failure would cause a change in the mechanical impedance, which in turn will change the electrical impedance obtained from the PZT bonded in the structure [7]–[9].

The PZT sensors are inexpensive, small, and lightweight, and require low power; they have a low sensitivity to temperature variations and have a linear response under low electric fields

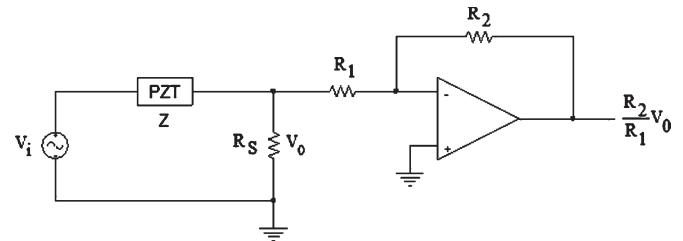


Fig. 1. Circuit proposed in [11] and [12] for PZT impedance assessment through a spectrum analyzer.

[10]. These properties make the E/M impedance technique efficient, accurate, and nondestructive for SHM application. However, the electrical impedance measurement is usually made with expensive, bulky, heavy, and high-power instruments, such as impedance analyzers (e.g., HP4194A and HP4192A), which are traditionally used in universities and research centers. These factors would certainly limit the use of large-scale E/M impedance-based techniques, such as in industrial and field applications or by universities with few financial resources.

Many researchers have proposed other systems to measure the electrical impedance in SHM applications. A method suggested in [11] and [12] uses a spectrum analyzer and a simple and inexpensive auxiliary circuit, as illustrated in Fig. 1. According to the authors, if the resistance R_S is considered small, then the PZT impedance is approximately given by $\dot{Z} = R_S \dot{V}_i / \dot{V}_o$, where $\dot{V}_i$ is the excitation signal, and $\dot{V}_o$ is the response signal, being both vectors. To obtain the impedance in the frequency domain, the discrete Fourier transforms (DFTs) of signals V_i and V_o (considered as discrete) are calculated by using a spectrum analyzer. In addition to the simplicity of the circuit and its low cost, this method possesses the following disadvantages: 1) The impedance is just an approximation that depends on the resistance R_S and tends to present a high discrepancy when the frequency increases because the PZT is predominantly capacitive. 2) If the resistance R_S is very small, then it is necessary to use a high-gain amplifier (R_2/R_1) that can reduce the bandwidth. 3) The spectrum analyzer is still an expensive instrument.

The impedance analyzer proposed in [13] only uses a resistor as an auxiliary circuit, and the signal response acquisition is performed by a data acquisition (DAQ) device, which is controlled by the software LabVIEW. It also seems that it is more accurate than the impedance analyzer suggested in [11] and [12]. However, the excitation signal is provided by an external function generator that needs an additional general purpose interface bus card. In addition to increasing the cost, these devices make the system less versatile.

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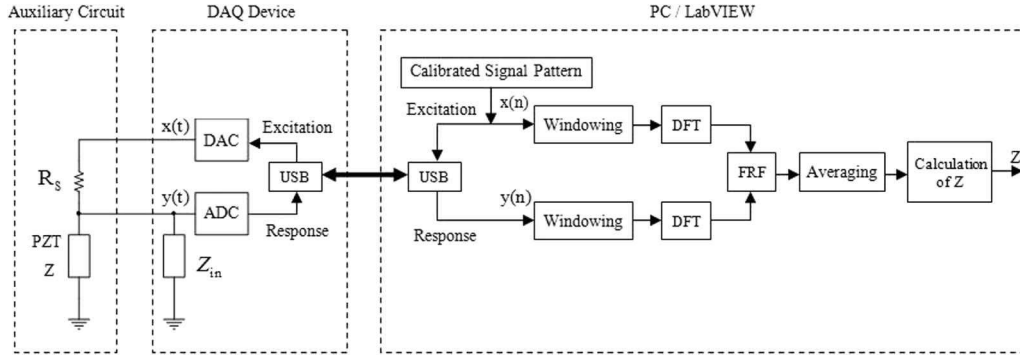


Fig. 2. Complete system proposed for PZT impedance measurement.

Recently, low-cost, low-power, and wireless impedance devices were developed. The system proposed in [14] is based on a DSP and uses a new algorithm that eliminates the use of analog-to-digital converters (ADCs) and digital-to-analog converters (DACs). However, the range and the frequency resolution are limited by the small memory space of the DSP. As a consequence, a conventional and expensive impedance analyzer is still necessary to determine the frequency range sensitive to structural damage. The devices proposed in [15] and [16] are based on AD5933, which is a miniaturized impedance-measuring chip made by Analog Devices. In [15], a statistical technique determines the main components of the impedance sensitive to structural damage, and in [16], the authors use a multiplexer for seven sensors, an autoranging capability for different impedance values, and more versatile sleep modes. Nevertheless, the AD5933 allows impedance measurements only going up to 100 kHz, which can limit many SHM applications. In [17], the authors proposed a complex circuit to design a self-contained SHM system, but the system was tested using a DAQ device connected to a PC running LabVIEW. In [18], the authors proposed the circuit developed in [17] to be used in a specific remote application replacing the DAQ device by other complex hardware based on a field-programmable gate array.

Finally, it seems that the most recent systems are being proposed for specific applications previously studied in the laboratory—even the ones based on wireless technology. Therefore, it is still necessary to use evaluation boards or dedicated printed circuit boards.

To eliminate some of the mentioned drawbacks, this paper presents a versatile, efficient, and inexpensive methodology, in which the accuracy of the impedance calculation does not depend on either the frequency range or the components used in auxiliary circuits. The proposed system successfully replaces an array of equipment used to develop SHM applications, such as impedance analyzers, spectrum analyzers, and signal generators. The main characteristics of the system are the following: 1) It only uses a DAQ device and a resistor to connect the PZT sensor. 2) It can generate different signals to excite the PZT. 3) There is no DAQ of the excitation signal during the measurements. 4) It is a very accurate impedance analyzer because it was validated in practical circuits with no SHM application. 5) It is very simple to implement different metrics to evaluate the structural damage. 6) Fi-

nally, the overall system can easily be adapted for real-time applications.

II. PROPOSED METHODOLOGY

The basic principle of operation is based on the frequency response function (FRF) obtained by DFT with regard to the excitation and response signals of the PZT circuit. Using the FRF and considering the detailed parameters of the analyzed circuit, it is possible to obtain the accurate PZT impedance. The complete system is depicted in Fig. 2. The hardware of the system consists of an inexpensive DAQ device model USB-6211 from National Instruments and a personal microcomputer (PC) laptop. The software was developed in LabVIEW, which is a graphical programming environment. The communication between the DAQ and the PC is performed through a universal serial bus (USB) port, which provides versatility to the system.

The DAQ device is multifunctional with analog channels based on 16-bit ADC and 16-bit DAC. The ADC has a differential input, which provides good noise immunity and allows the acquisition of small-amplitude signals (a few millivolts). This avoids the use of input amplifiers in the auxiliary circuit, which would affect the performance of the system at higher frequencies.

The excitation signal is generated by software, which allows the choice of a wide variety of signals, such as Chirp, sine, triangular, square, random, etc., and eliminates the DAQ process of the signal. This is possible because there is a calibration between the signal generated by software $x[n]$ and the signal $x(t)$ that reaches the auxiliary circuit, which is performed as follows: Before any measurement and after all the parameters have been defined, the DAC output and the ADC input are internally connected in the DAQ device, and the excitation signal is sampled and stored once to replace the signal generated by the software. Because the converters are not ideal devices, this procedure significantly reduces the discrepancies caused by the DAC. In addition, when computing the FRF, the ADC effects present in $x[n]$ and $y[n]$ (the response signal) will be compensated. This calibration procedure significantly increases the accuracy of the FRF. The system was verified for different signals, such as uniform white noise, Gaussian white noise, periodic random noise, and Chirp. The Chirp signal offered better results, and all the data presented in this paper were obtained with it. The Chirp signal makes a sweep from an

initial low frequency to a final high frequency, or *vice versa*. The sequence $x[n](n = 0, 1, \dots, N - 1)$ of a Chirp signal is given by

$$x[n] = A \cdot \sin \left(\frac{2\pi}{F_S} n \left(\frac{(f_2 - f_1)}{2N} n + f_1 \right) \right). \quad (1)$$

In (1), f_1 and f_2 represent the initial and final frequency values, respectively. A is the amplitude, N is the number of samples, and F_S is the sampling rate. According to [19], the PZT must be excited in a high-frequency range to enable the detection of small failures and to reduce the sensing region of each PZT. The used DAQ device allows a sampling rate of up to 250 kS/s and, as a consequence, a Nyquist frequency of 125 kHz. However, this frequency can easily be increased by choosing other high-speed DAQ devices. To compare the results with a conventional impedance analyzer, the PZT was excited from a dc level ($f_1 = 0$) up to $F_S/4$ ($f_2 = 62.5$ kHz). The relationship $f_2 N / 2 F_S$ provides the total number of cycles and should result in an integer number for the signal only containing integral cycles. A signal containing integral cycles presents little, or no, discontinuity. Thus, there is no spectral leakage, and it is not necessary to window it before computing its DFT. Setting up $N = 65\,536$ and replacing all the other parameters previously defined in (1), a Chirp signal with 8192 cycles is obtained, and its duration, which is given by the product $N \cdot T$ (where $T = 1/F_S$), is 262.144 ms.

To obtain the PZT impedance, it is necessary to determine the FRF from the auxiliary circuit, where the input is the excitation signal, and the output is the response signal. The first step is to compute the DFT of the input $x[n]$ and the output $y[n]$ by using fast Fourier transform (FFT) algorithms. This will result in $X[k]$ and $Y[k]$, which are complex sequences in the frequency domain. Each element of index k has a frequency corresponding to $k \cdot d_f$, where d_f is the spectral resolution defined by the relationship F_S/N . According to the previously defined values, d_f is approximately 3.815 Hz. Thus, the elements of $X[k]$ and $Y[k]$ have frequencies corresponding to $0, d_f, 2d_f, 3d_f, \dots, (N/4)d_f$, which is the highest frequency present in the Chirp signal.

From the excitation and response signals in the frequency domain, the FRF, which is denoted by $H[k]$, can be obtained as

$$H[k] = \frac{Y[k]}{X[k]}. \quad (2)$$

Depending on the environment and the length and type of the used cables, the system can be susceptible to noise caused by engines, reactors, and other devices. Therefore, although the used ADC converter has a differential input that provides good noise immunity, to get a high-quality FRF, it is proposed that an average between several excitation and response cycles is obtained. In this case, the averaged FRF is given by

$$\overline{H[k]} = \frac{\overline{Sxy[k]}}{\overline{Sxx[k]}}. \quad (3)$$

In (3), $\overline{Sxy[k]}$ and $\overline{Sxx[k]}$ represent the averages of the crossed power spectrum and the auto power spectrum of the

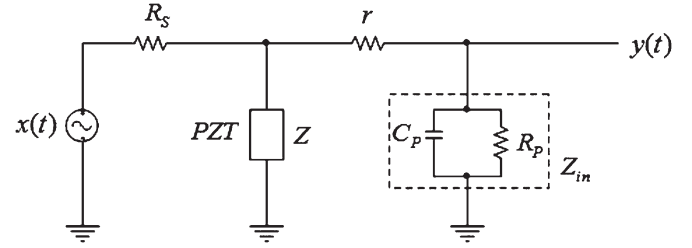


Fig. 3. Auxiliary circuit details for PZT excitation.

excitation signal, respectively. This is defined by

$$\overline{Sxy[k]}_{(i)} = \left(\overline{Sxy[k]}_{(i-1)} \cdot (i) + Sxy[k]_{(i)} \right) / (i+1) \quad (4a)$$

$$\overline{Sxx[k]}_{(i)} = \left(\overline{Sxx[k]}_{(i-1)} \cdot (i) + Sxx[k]_{(i)} \right) / (i+1). \quad (4b)$$

The power spectra of each cycle are given by

$$Sxy[k]_{(i)} = X[k]_{(i)}^* \cdot Y[k]_{(i)} \quad (5a)$$

$$Sxx[k]_{(i)} = |X[k]_{(i)}|^2. \quad (5b)$$

The index i represents the iteration and varies from 0 to $m - 1$, where m is the number of desired cycles. In (4a) and (4b), the averages of the power spectra of each measured cycle are weighted according to the corresponding iteration. However, at the end of m iterations, this calculation results in a simple average, where the power spectra are equally weighted. The advantage of this method is that the average is continuously calculated, and the m measures need not be stored, which simplifies the software and reduces the use of memory.

The averaged FRF obtained at the end of m cycles and its computation by using (3) reduce the influence of the possible noise present in the response signal $y(t)$. When computing the FFT of the excitation signal, it uses the signal stored at the beginning of the measurement process, which is not susceptible to any noise. Obviously, its continuous form $x(t)$ can suffer noise influence, but because it is a relatively high-amplitude signal compared with the response signal, the noise will not affect it. In addition to reducing the noise, the averaged FRF reduces the possible signal fluctuations caused by minor problems due to bad connections between the sensor and the system.

From the averaged FRF, it is possible to calculate the impedance Z of the PZT. The accuracy of this calculation depends on the parameters considered in the auxiliary circuit, which can be seen in detail in Fig. 3.

In the circuit of Fig. 3, R_S is a precision resistor that limits the current, r represents the resistance of the cable used to connect the DAQ device and the PZT sensor, and Z_{in} is the input impedance of the DAQ device. According to the manufacturer, this impedance consists of a resistance R_P of 10 G Ω in parallel with a capacitance C_P of 100 pF. Considering these parameters, the impedance $Z[k]$ of the PZT in function of the averaged FRF $\overline{H[k]}$ is given by

$$Z[k] = \frac{\overline{H[k]} \cdot R_S (r + Z_{in}[k])}{Z_{in}[k] - \overline{H[k]} (R_S + r + Z_{in}[k])}. \quad (6)$$

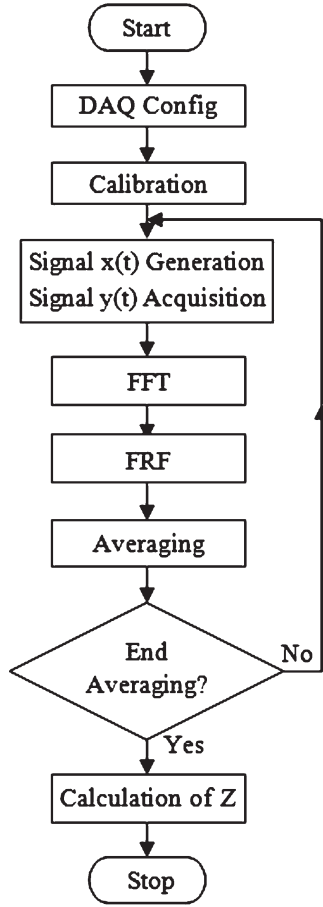


Fig. 4. Flowchart of the program.

In (6), the sequence $Z_{in}[k]$ is the input impedance of the DAQ device in function of the frequency, and it is calculated according to the following equation:

$$Z_{in}[k] = \left(\frac{1}{R_P} + j2\pi k d_f C_P \right)^{-1}. \quad (7)$$

The high value of the resistance R_P does not affect the results. However, the reactance of C_P becomes significantly low at higher frequencies and must be considered in all applications. The resistance r was not considered in this paper ($r = 0$), but it is important in cases where the structure to be monitored is far from the DAQ device and where the wires are too long. The resistor R_S should be chosen according to the maximum voltage desired in the PZT. It is recommended that this voltage not be very high, because the PZT tends to lose the linearity for intense electric fields. Using a Chirp signal with 5 V of amplitude, it was verified that a precision resistor of 10 K Ω ensures a voltage of below 1 V to the transducer for frequencies above 10 kHz.

The control of the DAQ device and all signal processing was performed using LabVIEW. The flowchart of the program is shown in Fig. 4.

In Fig. 4, the block DAQ Config sets the analog output for the generation of the excitation signal $x(t)$ and the analog input for the acquisition of the response signal $y(t)$. The other blocks represent the same significance that have already been

commented on and presented in the diagram in Fig. 2. The time spent on each iteration is about the same time required for the generation of the Chirp signal defined previously, i.e., the product $N \cdot T$. Therefore, an impedance measurement with m cycles of excitation/response is finished in $m \cdot N \cdot T$ seconds. Although the number of samples is high (65 536), the time needed for computing the FFT is very small when compared with the time for each iteration and, therefore, is not considered.

As shown in the next section, in addition to reducing the cost, the proposed methodology allows a quicker measure of impedance when compared with conventional instruments.

III. EXPERIMENTAL SETUP

The tests were performed using the USB-6211 DAQ device connected to the USB port of a notebook (Sempron 3600, 512 MB) running the operational system Windows and the LabVIEW package. Through a virtual panel, as seen in Fig. 5, it is possible to see the curves of impedance (modulus and the real and imaginary parts) and the set of parameters for the acquisition and generation of the signals, which were configured according to the values defined in the previous section.

To validate the proposed methodology, the impedances of two devices were analyzed: a series RC circuit with known impedance and a PZT patch of 20 mm $\times$ 20 mm $\times$ 0.2 mm bonded in a aluminum beam of 600 mm $\times$ 25 mm $\times$ 4 mm in free-free set, i.e., with the two ends suspended by elastics, as illustrated in Fig. 6. To simulate structural damage, a steel screw nut of 4-mm external diameter was bonded to the beam. Three tests were carried out with the screw nut positioned at 50, 200, and 400 mm from the PZT. The resistance in healthy condition (without the screw nut) was compared with the resistances of the three damaged conditions by employing the calculation of the root-mean-squared deviation (RMSD) damage metric [20].

In addition, the impedance of the beam was measured with the low-frequency impedance analyzer 4192A by Hewlett-Packard, and the results were compared with the those obtained with the proposed system. In this paper, all the measures were taken under a temperature of 29 $^{\circ}\text{C}$.

IV. RESULTS

A. RC Circuit

The impedance of an RC circuit consisting of a precision resistor of $(100 \pm 1) \Omega$ in series with a capacitor of $(100 \pm 5) \text{ nF}$ was measured with the proposed method. Fifty cycles of excitation and response signals were used to make up the averaged FRF shown in (3). The results are presented in Fig. 7, where it is possible to check and compare the measured and theoretical values of the impedance, resistance, and reactance.

The measured values of the impedance and reactance are in line with their theoretical values throughout the frequency range. The maximum errors for the impedance and the reactance were 6.1% and 5.6%, respectively, which are slightly above the accuracy of the capacitor ($\pm 5\%$).

There was a small discrepancy in the resistance measured in frequencies below 20 kHz. This may be because the values of

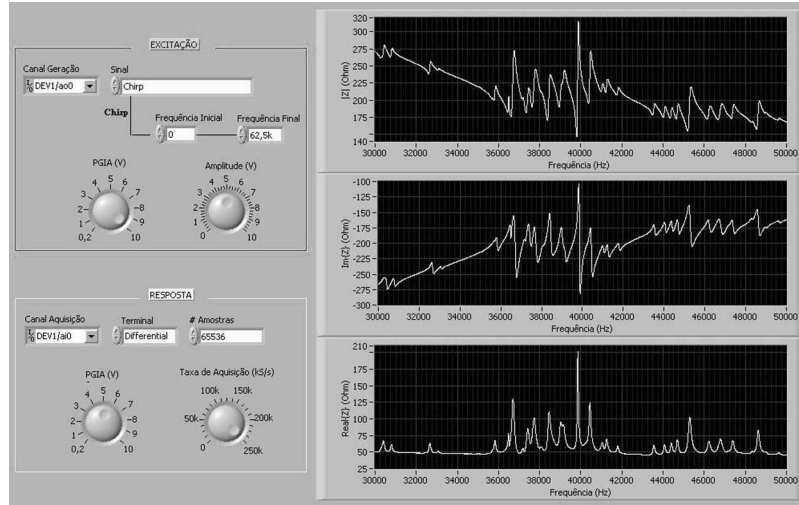


Fig. 5. Virtual panel of the proposed system.

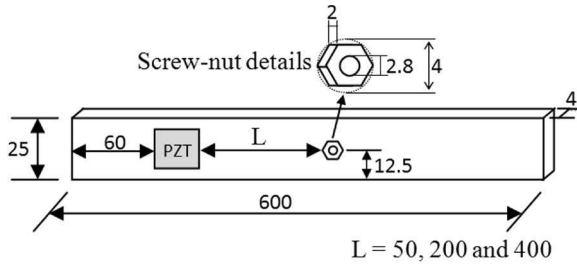


Fig. 6. Structure used for impedance measurement. Measures in millimeters.

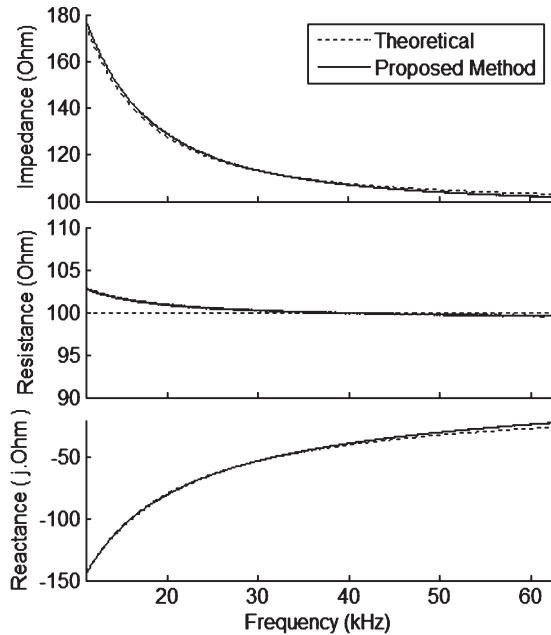


Fig. 7. Measured values and theoretical values of the impedance, resistance, and reactance of a series RC circuit.

the parameters defined in (6) are not exactly the same for the real system or due to other factors that are difficult to determine and measure, such as parasitic capacitance and parasitic inductance. However, this is not a problem in SHM systems, because the interest is at higher frequencies in which the proposed

method presented very accurate results. For frequencies above 20 kHz, the error is less than the accuracy of the resistor ($\pm 1\%$).

B. PZT Beam

The impedance of the PZT attached in the beam was measured with the proposed method and with the conventional impedance analyzer, taking care that the measures were carried out under the same conditions, i.e., same frequency range, resolution, and temperature. The measurement results and the discrepancies between them are shown in Fig. 8. The maximum and mean errors were 3.9% and 1.1%, respectively.

In Fig. 8, some oscillations can be observed in the impedance in some specified bands. These oscillations are due to the resonance that occurs in the natural frequencies of the structure. To detect structural damage, an impedance analyzer must be able to accurately identify changes in these frequencies.

Fig. 9 only shows the frequency range between 35 and 45 kHz, which allows a better comparison between measures of impedance, resistance, and reactance obtained with the proposed method and the conventional impedance analyzer. It can be verified that the shapes of the curves obtained with the proposed method are very similar with those obtained with the impedance analyzer. In addition, as verified in the impedance curve, the errors obtained in the resistance and reactance measures were very low (remaining below 5%).

Fig. 10 shows the measurement results of the resistance to the beam in a healthy condition (baseline) and with three damages simulated at a distance of 50, 200, and 400 mm from the PZT, as the procedure described in Section III. Within the evaluated frequency range, the natural frequencies between 35 and 42 kHz were shown to be more sensitive to the type of damage and were selected. The RMSD damage metric between the baseline and each measure in a damaged condition was computed. The results are shown in Table I, where baseline* denotes the results with the beam in a healthy condition; the small value obtained demonstrates the accuracy of the system. For a damage located at a distance of 400 mm from the PZT, the damage metric was 68.22 against 0.0012 in the healthy condition. These values

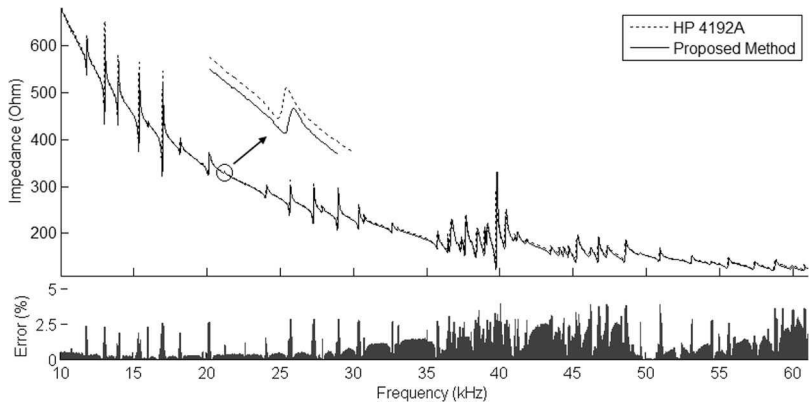


Fig. 8. Measures of the impedance of the set PZT beam obtained with the proposed method and with the conventional analyzer HP4192A.

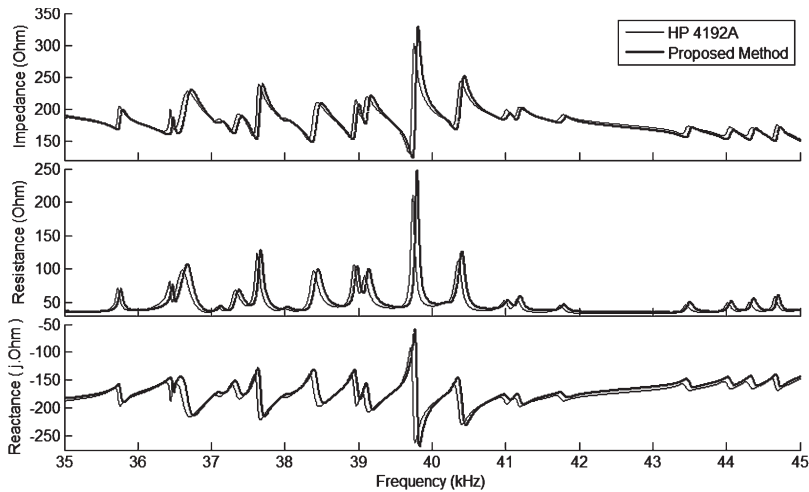


Fig. 9. Measures of impedance, resistance, and reactance of set PZT beam obtained with the proposed method and the conventional analyzer.

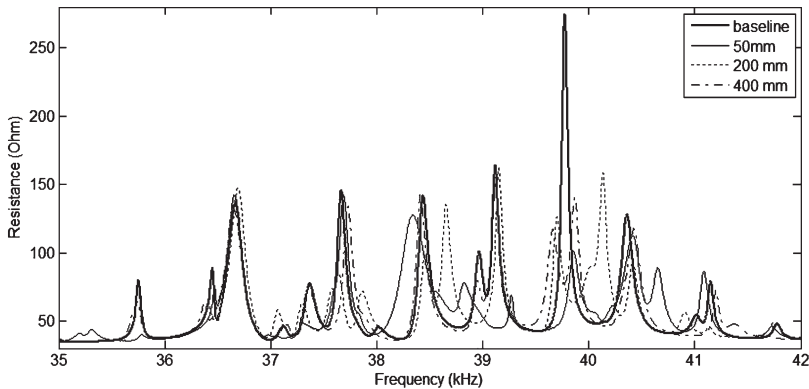


Fig. 10. Changes in resistance of the PZT beam caused by structural damages at distance of 50, 200, and 400 mm from the PZT and identified with the proposed method.

TABLE I DAMAGE METRIC RMSD	
Damage distance (mm)	RMSD
Baseline*	0.0012
50	438.52
200	437.51
400	68.22

can be changed according to the used PZT and the structure under test.

The test configuration was the same configuration used to evaluate the *RC* circuit, and the impedance measurement of the set PZT beam was performed with 50 cycles of excitation and response signal to make up the averaged FRF. The total time spent for each measure was less than 15 s. On the other hand, each measure acquired with the conventional impedance

analyzer, in the same frequency range and resolution, spent around 70 min. Clearly, HP4192A is an older model, and there are faster instruments on the market. However, such equipment are still expensive and impractical for most applications.

V. CONCLUSION

This paper has presented a new impedance measurement methodology for an SHM system based on the E/M impedance technique. The system offers precision, speed, low cost, and versatility, and may be used as a general impedance analyzer. The new system can be used to help new developments in the field of SHM, as well as help the transition of the E/M impedance-based technique from research to practice, where dedicated systems are implemented.

This paper used a DAQ device with a sampling rate limited to only 250 kS/s to verify the effectiveness of the proposed methodology. Devices with larger capacity and yet inexpensive can be used for the impedance analysis in higher frequencies with minor software modifications. Other programming platforms can be used in the place of LabVIEW, since an appropriate driver is used for the control of the device DAQ.

The methodology can easily be integrated into a system based on microcontroller DSP and, together with a wireless communication system, allows the E/M impedance technique to be used in the real-time monitoring of structures that are difficult to access or mobile, such as aircraft, for instance.

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